



Juan Vallejo

Backend Developer

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PROFESSIONAL SUMMARY

Backend Developer focused on designing scalable RESTful APIs and relational data models with Node.js, NestJS, Express, and ASP.NET Core. Experienced building modular, multi-role backend systems — spanning 25+ modules on some platforms — with PostgreSQL, MongoDB, Prisma, and Entity Framework Core, plus asynchronous processing with Redis and BullMQ. Implemented JWT-based authentication, role-based access control, and third-party integrations (Firebase Admin, WhatsApp Business API, AWS S3) in real production environments. Comfortable owning architecture decisions and deployments, carrying a Clean Architecture mindset across academic and professional projects alike.

TECHNICAL SKILLS

BACKEND & APIS

Node.js Express NestJS ASP.NET Core MVC RESTful APIs WebSockets Webhooks

PROGRAMMING LANGUAGES

JavaScript TypeScript Python C# SQL PHP

DATABASES & ORMS

PostgreSQL MongoDB Prisma Drizzle ORM Entity Framework Core

ASYNC & MESSAGING

Redis BullMQ Socket.io

ARCHITECTURE

Clean Architecture Modular / Domain-driven Design Multi-role Access Systems

SECURITY & AUTH

JWT bcrypt RBAC Passport.js Zod Validation

CLOUD & INTEGRATIONS

AWS S3 Firebase Admin SDK WhatsApp Business API

DEVOPS & TOOLS

Docker Git GitHub GitLab Postman Linux (Ubuntu)

AI-ASSISTED DEVELOPMENT

Claude Code

FEATURED PROJECTS

Urbanix — Residential Community Management SaaS

NestJS 11 Prisma PostgreSQL Redis WebSockets AWS S3 Firebase Admin

- ▶ Architected a NestJS backend with 25+ modules (finance, reservations, incidents, documents, notifications) serving a multi-role residential SaaS platform.
- ▶ Designed PostgreSQL schemas and JWT-secured REST APIs, plus real-time WebSocket channels and AWS S3-backed document storage.

Client project — Codeliance

Agentes — Social Media Automation Platform

NestJS · Prisma · PostgreSQL · Next.js

- ▶ Built the backend automation engine for a multi-platform social media task scheduler (likes, comments, follows) across Instagram, Twitter, TikTok, and Facebook.
- ▶ Designed the task/agent/account/log data model and execution pipeline.

Client project — Codeliance

DraAF — Medical Clinic Appointment Platform

Node.js · Express · TypeScript · MongoDB · Redis · BullMQ · WhatsApp Business API

- ▶ Built a complete Express/TypeScript API with Mongoose models, Zod-validated inputs, and JWT role-based auth (Admin/Patient).
- ▶ Designed an asynchronous BullMQ worker connected to the WhatsApp Business API webhook to automate appointment confirmations and cut no-shows.

ARVALA-SGPE — Poultry Farm Management System

ASP.NET Core 9 MVC · C# · PostgreSQL 17 · Entity Framework Core · Clean Architecture

- ▶ Implemented a Clean Architecture backend (Domain/Infrastructure/Services/Web layers) with ASP.NET Identity and EF Core over PostgreSQL to manage poultry production data.

PROFESSIONAL EXPERIENCE

Backend Developer — Codeliance

Feb 2025 – Present

- ▶ Design and automate backend workflows for administrative and documentary process management across multiple client platforms.
- ▶ Own backend architecture for production systems spanning residential management, sports club administration, and social automation.

Backend Developer — TechNova Solutions

Apr 2024 – Nov 2024

- ▶ Designed RESTful APIs with Node.js, Express, and Prisma over PostgreSQL for a residential-management SaaS application.
- ▶ Automated internal processes with Python and AutoHotKey, integrated Google Workspace workflows, and participated in architectural decisions and deployments.

EDUCATION

Computer Engineering (In Progress)

2023 – Present

Universidad Santiago de Cali, Colombia — Member, Comba Research & Development Incubator

CERTIFICATIONS

- ▶ **Databases:** Bases de Datos: Generalidades y Sistemas de Gestión — SENA (40h, 2025)
- ▶ **Programming:** C# — Ademass Training & Development · Python — Ademass Training & Development
- ▶ **Cybersecurity & Networking:** Ethical Hacker — Cisco Networking Academy (Verified) · Introduction to Cybersecurity — Cisco Networking Academy (Verified)

ACHIEVEMENTS & LEADERSHIP

- ▶ Vice President, IEEE Computer Society — Universidad Santiago de Cali (2025 – 2026)
- ▶ Active Member, ETH Cali — Web3 & blockchain hackathon participant (2026 – Present)

LANGUAGES

Spanish — Native

English — B2 (Upper Intermediate)